

Abhyudai Sood

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SKILLS SUMMARY

- **Languages:** C, C++, Java, Python, SQL, JavaScript
- **Libraries & Frameworks:** NumPy, Pandas, Matplotlib, Chart.js, Java Swing, Java Collections Framework, C++ STL
- **Web Technologies:** HTML and CSS
- **Tools/Platforms:** VS CODE, Git, GitHub, AutoCAD, Arduino IDE, Adobe Illustrator
- **Soft Skills:** Team Collaboration, Adaptability, Communication, Project Management, Leadership

PROJECTS

Smart Blood Donation & Emergency Matching Network | [GITHUB](#)

Jun 2026 – Jul 2026

- Developed a Java-based system for blood donor management and emergency request matching.
- Implemented HashMap, Nested HashMap, PriorityQueue, LinkedList, and Stack for efficient donor management and emergency processing.
- Optimized donor searches using blood-group and city indexing with O(1) average lookup and implemented priority-based emergency request handling.

Opti-Reach AI-driven rural medical health-camp planning | [GITHUB](#)

Mar 2026 – Apr 2026

- Created an AI-based rural healthcare optimization system using Linear Regression for village priority scoring.
- Applied a Genetic Algorithm for doctor–village–slot allocation, route optimization, and conflict reduction.
- Built an interactive dashboard with HTML, CSS, JavaScript, Leaflet.js, and Chart.js for route and data visualization

TraceX System Call simulation | [GITHUB](#)

Dec 2025

- Built a web-based Operating System simulator for demonstrating system calls and core OS operations.
- Simulated file handling, process management, and memory allocation through an interactive interface.
- Visualized system-call activity with Chart.js timeline, pie, and bar charts for analysis.

ACADEX – SPI Calculator| [GITHUB](#)

Jun 2025 - Jul 2025

- Developed core frontend components using HTML and CSS for the Acadex Student Performance Index website.
- Structured interfaces to present grades, CGPA, percentage, backlogs, and performance information.
- Optimized an intuitive and mobile-friendly user interface with clear navigation and organized performance data presentation

WHTS – We Help The Strays Website | [GITHUB](#)

Sep 2024 - Nov 2024

- Created a multi-page NGO website using HTML and CSS for organizational information and user interaction.
- Integrated donation, membership, navigation, forms, image, and video sections across the website.
- Organized website assets and applied custom CSS styling for consistent page presentation.

TRAININGS

Data Structures Fundamentals: Basics to Applications — LPU| [Certificate](#)

Jun 2026 - Jul 2026

- Learned DSA using Java, covering arrays, linked lists, stacks, queues, trees, searching, and sorting.
- Applied HashMap, PriorityQueue, LinkedList, and Stack to practical programming problems.
- Built the Smart Blood Donation Network using Java and Java Swing as a DSA based project.

Logic Building, Programming & Data Structures — LPU | [Certificate](#)

Jun 2025 - Jul 2025

- Learned C++ fundamentals through logic-building and problem-solving.
- Studied DSA concepts including arrays, linked lists, stacks, hashing, recursion, and dynamic programming.
- Developed the Acadex Student Performance Index using HTML and CSS as a team project.

RESEARCH

Automated Sliding Window Track Cleaning System — Patent Filed | 202511067767

Mar 2025 - Jul 2025

- Conceptualized an automated sliding-window track cleaning system using sensors and a motorized cleaning mechanism.
- Integrated automated cleaning, monitoring, and control mechanisms for efficient track maintenance.
- Filed an Indian patent application for the proposed automated cleaning system.

EDUCATION

- **Lovely Professional University**
Bachelor of Technology - Computer Science and Engineering
CGPA: 8.06
- **St. Edwards School**
Intermediate - PCM
Percentage: 72%

Punjab, India
Since Aug 2024

Shimla, Himachal Pradesh
Jul 2022 – May 2024